CYBERSHIELD AI

Intelligent Cyber Incident Detection and Safety Management Portal

Complete Project Details Document

SIH Problem Statement ID: SIH25183

Ministry of Defence, Government of India

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1. PROJECT OVERVIEW

CyberShield AI is a full-stack, AI-powered web portal designed for the Ministry of Defence to enable defence personnel to securely report cyber incidents and administrators to monitor, analyse, and respond to threats in real time. It was developed in response to Smart India Hackathon 2025 Problem Statement SIH25183 (AI-Enabled Cyber Incident & Safety Web Portal for Defence).

The portal features a zero-trust security architecture with Row-Level Security (RLS) at the database level, AI-powered threat analysis using Google Gemini, and real-time safety alert broadcasting. It is deployed on Vercel with automatic CI/CD from GitHub.

Live Deployment

- GitHub Repository: https://github.com/rathish1410-a11y/AI-cyber-safety-reportal-portal-
- Hosted on Vercel with automatic deployment on every push to main branch

2. COMPLETE TECHNOLOGY STACK

2.1 Frontend Technologies

Technology	Version	Used For
React	18.3.1	Component-based UI framework
TypeScript	5.5.3	Type-safe JavaScript development
Vite	5.4.2	Fast build tool and dev server
Tailwind CSS	3.4.1	Utility-first CSS styling
React Router	7.17.0	Client-side page navigation
Recharts	3.8.1	Interactive charts and graphs
Lucide React	0.344.0	Icon library for UI elements
React Leaflet	4.2.1	Interactive threat maps
Leaflet	1.9.4	Map rendering engine

2.2 Backend & Database

Technology	Used For
Supabase (supabase-js 2.57.4)	Backend-as-a-Service (PostgreSQL DB, Auth, REST API)
PostgreSQL	Relational database with Row-Level Security (RLS)
Supabase Auth	User authentication (email/password, session management
Supabase RLS	Zero-trust database-level data isolation policies

2.3 AI & Machine Learning

Technology	Used For
Google Gemini 2.5 Flash API	Primary AI engine for all intelligence features
Gemini - Risk Scoring	Analyses incident text and assigns risk score (0-100)
Gemini - CyberBot	AI chatbot for cybersecurity guidance
Gemini - Phishing Detector	URL/email analysis for phishing detection
Gemini - ATI Log Analyzer	Parses raw firewall/server logs for anomalies
3-Model Fallback Chain	gemini-2.5-flash > flash-latest > 2.0-flash

2.4 Deployment & DevOps

Technology	Used For
Vercel	CDN hosting and automatic deployment
GitHub	Source code repository and CI/CD trigger
ESLint	Code linting and quality enforcement
PostCSS + Autoprefixer	CSS post-processing for browser compatibility

3. DATABASE DESIGN

3.1 Database Tables

Table	Key Columns	Purpose
profiles	id, email, full_name, role	User profiles with role (citizen/admin)
incidents	id, title, type, severity, ai_risk_score	Cyber incident reports
alerts	id, title, content, severity, created_at	Safety alert broadcasts
ai_analyses	id, incident_id, risk_score, keywords	AI analysis results per incident

3.2 Row-Level Security (RLS) Policies

- Citizens can only SELECT and INSERT their own incidents (filtered by auth.uid())

- Admins can SELECT all incidents across all users

- Admins can UPDATE incident status and severity
- Citizens can only read their own profile
- Admins can read all profiles (via is_admin() SECURITY DEFINER function)
- All policies enforce zero-trust data isolation at the database level

4. COMPLETE FEATURE LIST

4.1 Authentication & Security

- Email/Password signup and login with Supabase Auth
- Automatic profile creation on signup with role assignment
- Forgot Password and Reset Password flow via email
- Protected routes: ProtectedRoute.tsx enforces role-based access
- Admin routes completely isolated from citizen routes
- Zero-Trust RLS: even if frontend is bypassed, database blocks unauthorized access

4.2 Defence Personnel (Citizen) Portal

- CitizenOverview.tsx - Dashboard with quick stats and navigation
- ReportIncident.tsx - Guided form with military categories (APT, Malware, Insider Threat, etc.)
- MyIncidents.tsx - View submitted incidents with status tracking and AI analysis results
- AIAssistantPage.tsx - Dedicated AI chatbot page for cybersecurity guidance
- AlertsPage.tsx - View safety alerts with loud audio siren notifications (Web Audio API)
- ThreatMap.tsx - Live interactive threat map showing user-reported incidents on a map
- ProfilePage.tsx - View and manage account profile
- ReportAnonymous.tsx - Submit incidents without revealing identity

4.3 Admin Command Centre

- AdminDashboard.tsx - KPI cards (total incidents, pending, critical, resolved) + recent incidents with reporter name
- AdminIncidents.tsx - Full searchable incident table with Reporter column, detail modals showing AI analysis
- AdminAnalytics.tsx - Interactive charts: incident trends (Area), severity distribution (Pie), category breakdown (Bar)
- AdminLogAnalyzer.tsx - ATI Log Analyzer: paste raw firewall/server/IDS logs, AI detects anomalies and suggests containment
- AdminAlerts.tsx - Create, edit, and delete safety alert broadcasts for all personnel

- ThreatMap.tsx (Admin view) - See ALL threats nationwide (RLS unlocks full data for admins)

4.4 AI-Powered Features

- AI Risk Scoring Engine: Analyses incident narrative and assigns 0-100 risk score with reasoning
- Risk categories: Low (0-25), Medium (26-50), High (51-75), Critical (76-100)
- CyberBot AI Chatbot: Floating widget with multi-turn conversation, India-specific guidance (helpline 1930, cybercrime.gov.in)
- Phishing/URL Detector: Analyses URLs and email text, returns verdict (Safe/Suspicious/Dangerous), threat score, and red flags
- ATI Log Analyzer: Parses unstructured security logs, identifies attack patterns, outputs military-grade containment protocols
- 3-Model Fallback Chain: If primary model fails (rate limit), auto-switches to backup models with retry delay

4.5 Safety Alert System

- Admins create alerts with title, content, and severity level
- Defence personnel see alerts in real-time on their Alerts page
- Loud, attention-grabbing synthesised audio siren using Web Audio API
- Exponential frequency sweep with 1.5-second duration for maximum attention

4.6 Live Threat Map

- Interactive Leaflet.js map showing reported cyber threats geographically
- Auto-detects user current location with blue dot indicator
- Citizens see only their own threats (RLS enforced)
- Admins see ALL threats nationwide (RLS bypass for admin role)
- Admin can double-click to manually add threat markers
- Location search functionality for navigation

4.7 Other Features

- Anonymous Reporting: Submit incidents without authentication
- Email typo validation on signup (warns for common mistakes)
- Cyberpunk-themed landing page tailored for defence context
- Responsive design for desktop and mobile
- Copy-to-clipboard report feature in Phishing Detector

5. COMPLETE PAGE LIST (19 Pages)

Page	File	Access
Landing Page	LandingPage.tsx	Public
Login	LoginPage.tsx	Public
Signup	SignupPage.tsx	Public
Forgot Password	ForgotPasswordPage.tsx	Public
Reset Password	ResetPasswordPage.tsx	Public
Anonymous Report	ReportAnonymous.tsx	Public
Citizen Dashboard	CitizenOverview.tsx	Citizen Only
Report Incident	ReportIncident.tsx	Citizen Only
My Incidents	MyIncidents.tsx	Citizen Only
AI Assistant	AIAssistantPage.tsx	Citizen Only
Safety Alerts	AlertsPage.tsx	Citizen Only
Phishing Detector	PhishingDetector.tsx	Citizen Only
Threat Map	ThreatMap.tsx	Both (filtered)
Profile	ProfilePage.tsx	Both
Admin Dashboard	AdminDashboard.tsx	Admin Only
Admin Incidents	AdminIncidents.tsx	Admin Only
Admin Analytics	AdminAnalytics.tsx	Admin Only
Admin Log Analyzer	AdminLogAnalyzer.tsx	Admin Only
Admin Alerts	AdminAlerts.tsx	Admin Only

6. PERFORMANCE METRICS

Metric	Value
Page Load Time	< 2 seconds (Vercel CDN)
Supabase Query Latency	50 - 150 ms average
Gemini AI Response Time	1 - 3 seconds
Production Bundle Size	~450 KB (gzipped)
Lighthouse Performance	85+
Lighthouse Accessibility	90+
RLS Data Isolation	100% enforced
AI Availability	99%+ (3-model fallback)

7. PROJECT FILE STRUCTURE

```
AI-cyber-safety-reportal-portal-/
```

```
|-- src/
|   |-- components/        (Reusable UI components)
|   |   |-- ChatBot.tsx    (Floating AI chatbot widget)
|   |   |-- ProtectedRoute.tsx (Role-based route guard)
|   |   |-- AdminSidebar.tsx   (Admin navigation sidebar)
|   |   |-- CitizenSidebar.tsx (Citizen navigation sidebar)
|   |-- context/
|   |   |-- AuthContext.tsx (Authentication state provider)
|   |-- lib/
|   |   |-- supabase.ts    (Supabase client configuration)
|   |-- pages/
|   |   |-- LandingPage.tsx, LoginPage.tsx, SignupPage.tsx ...
|   |   |-- admin/         (Admin-only pages)
|   |   |-- citizen/       (Citizen-only pages)
|   |   |-- tools/         (PhishingDetector)
|   |-- App.tsx            (Main app with routing)
|   |-- main.tsx           (Entry point)
|-- package.json           (Dependencies)
|-- tailwind.config.js     (Tailwind CSS configuration)
|-- vite.config.ts         (Vite build configuration)
|-- vercel.json            (Vercel deployment config)
|-- index.html             (HTML entry point)
```

8. QUICK SUMMARY

Item	Detail
Project Name	CyberShield AI
SIH Problem ID	SIH25183 (Ministry of Defence)
Student	RATHISH R (922525106247)
College	VSB Engineering College, Karur
Department	Electronics & Communication Engineering
Guide	Ms. AYSHA, Assistant Professor
Academic Year	2025-2029 (II Year: 2025-2026)
Frontend	React 18 + TypeScript + Tailwind CSS
Backend	Supabase (PostgreSQL + Auth + RLS)
AI Engine	Google Gemini 2.5 Flash API
Deployment	Vercel (CDN) + GitHub (CI/CD)
Total Pages	19 pages across 3 portals
AI Features	4 (Risk Scoring, Chatbot, Phishing, ATI)
Security	Zero-Trust RLS + RBAC
Build Tool	Vite 5.4

END OF DOCUMENT

CyberShield AI: Intelligent Cyber Incident Detection and Safety Management Portal

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